

Lecture notes, week 9

1 Compactness

A subset of $\mathbb{R}^n$ is compact exactly when it is closed and bounded; this is the Heine–Borel theorem, proved in [?].

2 Exercises

1. Show that a closed subset of a compact set is compact.
2. Show that a continuous image of a compact set is compact.
3. Give an open cover of $(0, 1)$ with no finite subcover.
4. Deduce from Section 1 that a continuous function on $[a, b]$ is bounded.